

Highlights

Citations **2,531 citations, h-index 24** (Google Scholar, Sep. 2026), rising every year since 2019; most-cited first-author papers: Single-Image HDR Reconstruction (CVPR 2020, 411), Robust Dynamic Radiance Fields (CVPR 2023, 337), Deep Video Frame Interpolation (AAAI 2019 oral, 216)

Publications **54 peer-reviewed conference papers** since joining NYCU in 2023, **43 of them at top-tier conferences** (CVPR, ICCV, ECCV, NeurIPS, ICLR, AAAI, SIGGRAPH, SIGGRAPH Asia, IROS, CoRL); **3 journal articles** (IEEE TPAMI, IJCV, ACM TOG); **70 peer-reviewed papers** in total

Patents **4 U.S. patent applications with MediaTek** (2025–2026) and **4 U.S. patent applications with NVIDIA** (2025–2026), all from industry–academia projects led at NYCU

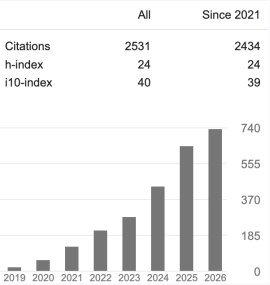
Funding **NT\$38.4M** as PI/co-PI since 2023, incl. **MOE Yushan Young Fellow** (NT\$15M, 2023–2028), **NSTC 2030 Emerging Young Scholar** (NT\$4.5M, 2024–2028), two consecutive MediaTek MARC projects (NT\$7M, 2023–2029), four consecutive NVIDIA TRDC projects (NT\$4.1M, 2023–2027), **Google Research Scholar Award** (one of the first two recipients at Taiwanese institutions, NT\$1.1M, 2024–2025)

Honors **Yushan Young Fellow** (Ministry of Education), **2030 Emerging Young Scholar** (NSTC), **Google Research Scholar Award**, **K. T. Li Young Researcher Award**, NYCU Excellent Teaching Award and Outstanding Mentor Award

Service **Area Chair**: CVPR 2027, NeurIPS 2025–2026, ICLR 2027, AAAI 2027, ACCV 2026; **Publication Chair**, ACCV 2024; co-organizer of Taiwan Night at CVPR and SIGGRAPH

Industry **7.5 years at MediaTek** on ISP and computer vision algorithms for mobile SoCs; 6 months at **Meta Reality Labs Research** on 4D reconstruction

Cited by



Research Interests

I work on *image/video processing*, *computer vision*, and *computational photography*, particularly on essential problems requiring *machine learning* with insights from *geometry* and *domain-specific knowledge*.

- Image and video synthesis and generation
- 3D from multi-view and sensors
- Low-level vision
- Vision + graphics

Work Experience

- Feb. 2026 – **Department of Computer Science, National Yang Ming Chiao Tung University.**
Associate Professor
- Feb. 2023 – **Department of Computer Science, National Yang Ming Chiao Tung University.**
Feb. 2026 *Assistant Professor*
- May 2022 – **Meta Reality Labs Research.**
Oct. 2022 *Research Scientist Intern*
- Jun. 2017 – **MediaTek Inc.**
Feb. 2022 *Senior Algorithm Development Engineer*
- Sep. 2014 – **MediaTek Inc.**
Jun. 2017 *Algorithm Development Engineer*

Selected Honors and Awards

- Sep. 2026 **Outstanding Mentor Award**, NYCU.

- Aug. 2026 **Excellent Teaching Award**, NYCU.
- Mar. 2025 **K. T. Li Young Researcher Award**, Institute of Information & Computing Machinery (IICM).
- Dec. 2024 **Outstanding Contribution Award for Young CS Scholars**, *College of CS*, NYCU.
- June 2024 **NSTC 2030 Emerging Young Scholar**, *National Science and Technology Council*, R.O.C. (Taiwan).
- May 2024 **CVPR 2024 Outstanding Reviewer**.
- April 2024 **Google Research Scholar Award**, *Google Research*.
One of the first two recipients at Taiwanese institutions.
- Aug. 2023 **Google Research Grant**, Google University Relations Programs.
- Aug. 2023 **Best Ph.D. Thesis Award**, The Chinese Image Processing and Pattern Recognition Society (IPPR).
- Jul. 2023 **Yushan Young Fellow Program**, *Ministry of Education*, R.O.C. (Taiwan).
- Nov. 2013 **Academic Achievement Award**, *Institute of Electronics*, NCTU.
- May. 2013 **Academic Achievement Award**, *Institute of Electronics*, NCTU.

Research Grants

- Aug. 2023 – **Yushan Young Fellow Program**, *Ministry of Education, Taiwan*, PI, NT\$15,000,000.
- Jul. 2028 NT\$3M per year for five years: NT\$1.5M salary supplement and NT\$1.5M research and administrative support.
- Aug. 2024 – **2030 Cross-Generation Young Scholars Program – Emerging Young Scholar: Enhanced ClimateNeRF: Integrating 3D Gaussian Splatting for Real-Time Dynamic Weather Simulations and Advanced Environmental Interactions in Neural Rendering**, *National Science and Technology Council (NSTC)*, PI, NT\$4,520,000.
- Jul. 2028
- May 2023 – **Research Project for Newly Hired Faculty: Data Augmentation for Neural Radiance Fields in Few-Shot Learning**, *National Science and Technology Council (NSTC)*, PI, NT\$1,851,000.
- Apr. 2025
- Jul. 2026 – **Efficient Continuous Spatio-Temporal Video Super-Resolution via Diffusion Transformers and Implicit Neural Representations**, *MediaTek Advanced Research Center (MARC), industry-academia program*, PI, NT\$3,500,000.
- Dec. 2029
Outputs: Stream-DiffVSR (ECCV 2026 spotlight), RL-AWB (ECCV 2026), PriSM (ECCV 2026); two U.S. patent filings.
- Jul. 2023 – **Towards Interpretable, Generalizable, and Controllable Video Restoration**, *MediaTek Advanced Research Center (MARC), industry-academia program*, PI, NT\$3,500,000.
- Dec. 2026
Outputs: DiffIR2VR-Zero, GCC (CVPR 2025), HDR Reconstruction Boosting (WACV 2026); two U.S. patent filings.
- Aug. 2026 – **Real-Time Large-Scale View Synthesis and Physics-Aware Generative World Models via Test-Time Training**, *NVIDIA Taiwan Research & Development Center (TRDC), industry-academia program*, PI, NT\$1,200,000.
- Jun. 2027
Outputs: TTB.
- Aug. 2025 – **Pose-Free 3D Reconstruction and Tracking for Dynamic Scenes**, *NVIDIA Taiwan Research & Development Center (TRDC), industry-academia program*, PI, NT\$1,200,000.
- Jul. 2026
Outputs: LongSplat (ICCV 2025), Scal3R (ECCV 2026), 3AM; three U.S. patent filings.
- Aug. 2024 – **3D-Aware Tracking and Pose-Free Dynamic Scene Reconstruction with Diffusion-Guided Decomposition**, *NVIDIA Taiwan Research & Development Center (TRDC), industry-academia program*, PI, NT\$1,200,000.
- Jul. 2025
Output: AuraFusion360 (CVPR 2025); one U.S. patent filing.
- Jul. 2023 – **Deep Cross-domain Learning for Computer Vision Applications**, *NVIDIA Taiwan Research & Development Center (TRDC), industry-academia program*, Co-PI with Yen-Yu Lin, NT\$500,000.
- Jun. 2024
Output: Image-Text Co-Decomposition (CVPR 2024).
- 2024 **Google Research Scholar Award**, *Google Research*, PI, US\$35,000.
One of the first two recipients at Taiwanese institutions.
- 2023 **Google Research Grant**, *Google University Relations Programs*, PI, US\$30,000.
- May 2026 – **Toward Next-Generation AI Program, Sub-project 1: Unified Scene Editing with Multi-View Diffusion Models and Reference Sheets**, *National Science and Technology Council (NSTC), AI Center of Excellence program*, Co-PI (sub-project co-led with Wen-Hsiao Peng), NT\$1,400,000.
- Apr. 2027

- Apr. 2026 – **Beyond Infinigen: Toward an Open-Ended Procedural World**, *Shanda AI Research Tokyo (Alaya Lab)*, *industry-academia program*, PI, NT\$1,120,000.
 Mar. 2027
- Jan. 2024 – **2D-to-3D Scene Reconstruction and Simulation**, *Industrial Technology Research Institute (ITRI)*,
 Dec. 2024 *industry-academia program*, PI, NT\$1,000,000.
- Jan. 2024 – **AI Core Lab for Coronary Angiography: Automated Multimodal 3D Image Interpretation, Coronary Perfusion Analysis, and CAD Prognosis**, *National Taiwan University Hospital (NTUH)*, PI, NT\$400,000.
 Dec. 2024 Output: DeNVeR (CVPR 2025).
- Total NT\$38.4M in external research funding as PI or co-PI since 2023.

Patents

- 2025 **Generative Color Constancy**, *U.S. Patent Application, MediaTek Inc.*, filed May 2025.
 Inventors: Chen-Wei Chang, Cheng-De Fan, Chia-Che Chang, Yi-Chen Lo, Jiun-Long Huang, Yu-Lun Liu. Based on GCC (CVPR 2025); MediaTek MARC industry-academia project.
- 2026 **System and Method for HDR Reconstruction**, *U.S. Patent Application, MediaTek Inc.*, filed Mar. 2026.
 Inventors: Yo-Tin Lin, Su-Kai Chen, Hou-Ning Hu, Yen-Yu Lin, Yu-Lun Liu. Based on HDR Reconstruction Boosting with Training-Free and Exposure-Consistent Diffusion (WACV 2026); MediaTek MARC industry-academia project.
- 2025 **Unsupervised Domain Adaptation for Facial Landmark Detection**, *U.S. Provisional Patent Application, MediaTek Inc.*, filed Nov. 17, 2025.
 Inventors: Yen-Yu Lin, Chieh-Yu Yang, Yu-Lun Liu, Su-Kai Chen, Hou-Ning Hu. Based on PriSM (ECCV 2026); MediaTek MARC industry-academia project.
- 2025 **Auto White Balance Correction for Low-Light Scenes via Reinforcement Learning**, *U.S. Provisional Patent Application, MediaTek Inc.*, filed Nov. 25, 2025.
 Inventors: Yuan-Kang Lee, Kuan-Lin Chen, Chia-Che Chang, Yu-Lun Liu. Based on RL-AWB (ECCV 2026); MediaTek MARC industry-academia project.
- 2025–2026 **Augmented Unseen Region Alignment for Reference-Based Scene Inpainting**, *U.S. Patent Application No. 19/543,594, NVIDIA Corporation*, filed Feb. 18, 2026; claims priority to U.S. Provisional Application No. 63/760,056, filed Feb. 18, 2025.
 Inventors: Min-Hung Chen, Chung-Ho Wu, Yang-Jung Chen, Ying-Huan Chen, Yen-Yu Lin, Yu-Lun Liu. Based on AuraFusion360 (CVPR 2025).
- 2025 **Robust Unposed Three-Dimensional Gaussian Splatting for Casual Long Videos Using Octree Anchor Formations**, *U.S. Patent Application, NVIDIA Corporation*, filed Sep. 2025; claims priority to U.S. Provisional Application No. 63/788,488, filed Apr. 14, 2025.
 Inventors: Fu-En Yang, Min-Hung Chen, Cheng Sun, Chin-Yang Lin, Yen-Yu Lin, Yu-Lun Liu. Based on LongSplat (ICCV 2025).
- 2026 **Relative Pose Querying for Online 3D Reconstruction**, *U.S. Provisional Patent Application No. 64/002,502, NVIDIA Corporation*, filed Mar. 11, 2026.
 Inventors: Cheng Sun, Chin-Yang Lin, Fu-En Yang, Min-Hung Chen, Yang-Che Sun, Yen-Yu Lin, Yu-Lun Liu. Based on Scal3R (ECCV 2026).
- 2025 **Multi-View Consistent Video Object Segmentation**, *U.S. Provisional Patent Application No. 63/943,823, NVIDIA Corporation*, filed Dec. 18, 2025; non-provisional application in preparation.
 Inventors: Yang-Che Sun, Cheng Sun, Chin-Yang Lin, Fu-En Yang, Min-Hung Chen, Yen-Yu Lin, Yu-Lun Liu. Based on 3AM (2026).

Student Honors

- Aug. 2026 **Best Master Thesis Award**, The Chinese Image Processing and Pattern Recognition Society (IPPR).
- Aug. 2026 **Excellent Master Thesis Award**, The Chinese Image Processing and Pattern Recognition Society (IPPR).
- July 2026 **1st place of the NYCU CS Undergraduate Research Competition**, CS, NYCU.
- July 2026 **3rd place of the NYCU CS Undergraduate Research Competition**, CS, NYCU.
- July 2026 **College Student Research Creativity Award**, National Science and Technology Council (NSTC).
- June 2026 **Hon Hai Technology Award**, Hon Hai Education Foundation.
- Jan. 2026 **3rd place of the NYCU CS Undergraduate Research Competition**, CS, NYCU.

- Dec. 2025 **Best Master Thesis Award**, Taiwanese Association for Artificial Intelligence (TAAI).
- Dec. 2025 **Excellent Master Thesis Award**, Taiwanese Association for Artificial Intelligence (TAAI).
- July 2025 **3rd place of the NYCU CS Undergraduate Research Competition**, CS, NYCU.
- July 2025 **2nd place of the NYCU CS Undergraduate Research Competition**, CS, NYCU.
- Nov. 2024 **Excellent Master Thesis Award**, Taiwanese Association for Artificial Intelligence (TAAI).
- Aug. 2024 **Excellent Master Thesis Award**, The Chinese Image Processing and Pattern Recognition Society (IPPR).
- July 2024 **3rd place of the NYCU CS Undergraduate Research Competition**, CS, NYCU.
- July 2024 **2nd place of the NYCU CS Undergraduate Research Competition**, CS, NYCU.
- Jan. 2024 **1st place of the NYCU CS Undergraduate Research Competition**, CS, NYCU.

Education

- 2017–2022 **Ph.D.**, *National Taiwan University*, Taipei, Taiwan.
 Department of Computer Science and Information Engineering, CMLAB
 Thesis: Restoring and Enhancing Images and Videos by Combining Modeling and Learning
 Advisor: Yung-Yu Chuang [link](#)
- 2012–2014 **M.S.**, *National Chiao-Tung University*, Hsinchu, Taiwan.
 Institute of Electronics, CommLab
 Thesis: Wide-Angle Virtual View Synthesis with Depth-Based Background Modeling
 Advisor: Hsueh-Ming Hang [link](#)
- 2008–2012 **B.S.**, *National Chiao-Tung University*, Hsinchu, Taiwan.
 Department of Electronics Engineering

Publications [\(Google Scholar profile\)](#)

Journal papers

- ACM TOG 2026 **DiffPhD: A Unified Differentiable Solver for Projective Heterogeneous Materials in Elastodynamics with Contact-Rich GPU-Acceleration.**
 Shih-Yu Lai, Sung-Han Tien, Jui-I Huang, Yen-Chen Tseng, Yi-Ting Chiu, Siyuan Luo, Ziqiu Zeng, Fan Shi, Peter Yichen Chen, Tiantian Liu, [Yu-Lun Liu](#), Bing-Yu Chen
 ACM Transactions on Graphics (SIGGRAPH Asia), 2026 [paper](#) [project](#)
- IJCV 2024 **Portrait Distortion Correction with Perspective-Aware 3D GANs.**
 Zhixiang Wang, [Yu-Lun Liu](#), Jia-Bin Huang, Shin'ichi Satoh, Sizhuo Ma, Guru Krishnan, Jian Wang
 International Journal of Computer Vision, 2024 [paper](#) [project](#)
- IEEE TPAMI 2021 **Learning to See Through Obstructions with Layered Decomposition.**
[Yu-Lun Liu](#), Wei-Sheng Lai, Ming-Hsuan Yang, Yung-Yu Chuang, and Jia-Bin Huang
 IEEE Transactions on Pattern Analysis and Machine Intelligence, 2021 [paper](#) [project](#)

Conference papers

- ICIP 2026 **IC-4DGS: Illumination-Compensated 4D Gaussian Splatting Under Photometric Variations.**
 He-Bi Yang, Ming-Zhe He, Cheng-Wei Yang, Jui-Chiu Chiang, [Yu-Lun Liu](#), Wen-Hsiao Peng
 IEEE International Conference on Image Processing, 2026 [paper](#)
- ICIP 2026 **VLM-Dreamer: VLM-Imagined Bi-Directional Inpainting for Single-Image 360 Scene Generation.**
 Ting-Wei Huang, Fu-En Yang, Min-Hung Chen, Yen-Yu Lin, [Yu-Lun Liu](#)
 IEEE International Conference on Image Processing, 2026 [paper](#) [project](#)
- ECCV 2026 **PriSM: Parsing and Style-Mixed Consistency for Unsupervised Domain Adaptation in Facial Landmark Detection.**
 Chieh-Yu Yang, Hou-Ning Hu, Sykai Chen, [Yu-Lun Liu](#), Yen-Yu Lin
 Proceedings of European Conference on Computer Vision, 2026 [paper](#) [project](#)
- ECCV 2026 **Scal3R: Learning Efficient Multi-Relative Pose Query for Scalable Online 3D Reconstruction.**
 Chin-Yang Lin, Yang-Che Sun, Cheng Sun, Fu-En Yang, Min-Hung Chen, Yen-Yu Lin, Wei-Chen Chiu, [Yu-Lun Liu](#)
 Proceedings of European Conference on Computer Vision, 2026 [paper](#) [project](#)
- ECCV 2026 **JanusMesh: Fast and Zero-Shot 3D Visual Illusion Generation via Cross-Space Denoising.**
 Siang-Ling Zhang, Huai-Hsun Cheng, Tsung-Ju Yang, [Yu-Lun Liu](#)
 Proceedings of European Conference on Computer Vision, 2026 [paper](#) [project](#)

- ECCV 2026 **DF3DV-1K: A Large-Scale Dataset and Benchmark for Distractor-Free Novel View Synthesis.**
Cheng-You Lu, Yi-Shan Hung, Wei-Ling Chi, Hao-Ping Wang, Charlie Li-Ting Tsai, Yu-Cheng Chang, [Yu-Lun Liu](#), Thomas Do, Chin-Teng Lin
Proceedings of European Conference on Computer Vision, 2026 [i paper](#) [i project](#)
- ECCV 2026 **RL-AWB: Deep Reinforcement Learning for Auto White Balance Correction in Low-Light Night-time Scenes.**
Yuan-Kang Lee, Kuan-Lin Chen, Chia-Che Chang, [Yu-Lun Liu](#)
Proceedings of European Conference on Computer Vision, 2026 [i paper](#) [i project](#)
- ECCV 2026 **Stream-DiffVSR: Low-Latency Streamable Video Super-Resolution via Auto-Regressive Diffusion.**
Hau-Shiang Shiu, Chin-Yang Lin, Zhixiang Wang, Chi-Wei Hsiao, Po-Fan Yu, Yu-Chih Chen, [Yu-Lun Liu](#)
Proceedings of European Conference on Computer Vision, 2026 [i paper](#) [i project](#)
Spotlight
- ECCV 2026 **Skyfall-GS: Synthesizing Immersive 3D Urban Scenes from Satellite Imagery.**
Jie-Ying Lee, Yi-Ruei Liu, Shr-Ruei Tsai, Wei-Cheng Chang, Chung-Ho Wu, Jiewen Chan, Zhenjun Zhao, Chieh Hubert Lin, [Yu-Lun Liu](#)
Proceedings of European Conference on Computer Vision, 2026 [i paper](#) [i project](#)
- SIGGRAPH 2026 **LongE2V: Long-Horizon Event-based Video Reconstruction, Prediction, and Frame Interpolation with Video Diffusion Models.**
Cheng-De Fan, Chun-Wei Tuan Mu, Chen-Wei Chang, Chin-Yang Lin, Kun-Ru Wu, Yu-Chee Tseng, [Yu-Lun Liu](#)
ACM SIGGRAPH 2026 Conference Papers, 2026 [i paper](#) [i project](#)
- SIGGRAPH 2026 **Stroke of Surprise: Progressive Semantic Illusions in Vector Sketching.**
Huai-Hsun Cheng, Siang-Ling Zhang, [Yu-Lun Liu](#)
ACM SIGGRAPH 2026 Conference Papers, 2026 [i paper](#) [i project](#)
- SIGGRAPH 2026 **VecSet-Edit: Unleashing Pre-trained LRM for Mesh Editing from Single Image.**
Teng-Fang Hsiao, Bo-Kai Ruan, [Yu-Lun Liu](#), Hong-Han Shuai
ACM SIGGRAPH 2026 Conference Papers, 2026 [i paper](#) [i project](#)
- CVPR Findings 2026 **AdaGaR: Adaptive Gabor Representation for Dynamic Scene Reconstruction.**
Jiewen Chan, Zhenjun Zhao, [Yu-Lun Liu](#)
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026 [i paper](#) [i project](#)
- CVPR Findings 2026 **Gen-n-Val: Agentic Image Data Generation and Validation.**
Jing-En Huang, I-Sheng Fang, Tzuhsuan Huang, [Yu-Lun Liu](#), Chih-Yu Wang, Jun-Cheng Chen
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026 [i paper](#) [i project](#)
- CVPR 2026 **Pantheon360: Taming Digital Twin Generation via 3D-Aware 360° Video Diffusion.**
Ting-Hsuan Chen*, Ying-Huan Chen*, Tao Tu, Jie-Ying Lee, Cho-Ying Wu, Fangzhou Lin, Hengyuan Zhang, David Paz, Xinyu Huang, Yuliang Guo†, [Yu-Lun Liu](#)†, Yue Wang†, Liu Ren
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026 [i paper](#) [i project](#)
- CVPR 2026 **Reflection Separation from a Single Image via Joint Latent Diffusion.**
Zheng-Hui Huang, Zhixiang Wang, [Yu-Lun Liu](#), Yung-Yu Chuang
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026 [i paper](#) [i project](#)
- CVPR 2026 **PhaSR: Generalized Image Shadow Removal with Physically Aligned Priors.**
Chia-Ming Lee, Yu-Fan Lin, Yu-Jou Hsiao, Jin-Hui Jiang, [Yu-Lun Liu](#), Chih-Chung Hsu
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026 [i paper](#) [i project](#)
- CVPR 2026 **ReflexSplit: Single Image Reflection Separation via Layer Fusion-Separation.**
Chia-Ming Lee, Yu-Fan Lin, Jin-Hui Jiang, Yu-Jou Hsiao, Chih-Chung Hsu, [Yu-Lun Liu](#)
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026 [i paper](#) [i project](#)
- CVPR 2026 **Voxify3D: Pixel Art Meets Volumetric Rendering.**
Yi-Chuan Huang, Jiewen Chan, Hao-Jen Chien, [Yu-Lun Liu](#)
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026 [i paper](#) [i project](#)
- WACV 2026 **PS3: Part level instance segmentation in 3D.**
Hong-Xuan Yen, Chiamin Chen, Yanqing Wang, [Yu-Lun Liu](#), Min Sun
Proceedings of IEEE/CVF Winter Conference on Applications of Computer Vision, 2026 [i paper](#)
- WACV 2026 **HDR Reconstruction Boosting with Training-Free and Exposure-Consistent Diffusion.**
Yo-Tin Lin, Sykai Chen, Hou-Ning Hu, Yen-Yu Lin, [Yu-Lun Liu](#)
Proceedings of IEEE/CVF Winter Conference on Applications of Computer Vision, 2026 [i paper](#) [i project](#)

- WACV 2026 **TED-4DGS: Temporally Activated and Embedding-based Deformation for 4DGS Compression.**
Cheng-Yuan Ho, He-Bi Yang, Jui-Chiu Chiang, [Yu-Lun Liu](#), Wen-Hsiao Peng
Proceedings of IEEE/CVF Winter Conference on Applications of Computer Vision, 2026 [i](#) [paper](#) [i](#) [project](#)
- WACV 2026 **Splannequin: Freezing Monocular Mannequin-Challenge Footage with Dual-Detection Splatting.**
Hao-Jen Chien, Yi-Chuan Huang, Chung-Ho Wu, Wei-Lun Chao, [Yu-Lun Liu](#)
Proceedings of IEEE/CVF Winter Conference on Applications of Computer Vision, 2026 [i](#) [paper](#) [i](#) [project](#)
- SIGGRAPH Asia 2025 **Prior-Enhanced Gaussian Splatting for Dynamic Scene Reconstruction from Casual Video.**
Meng-Li Shih, Ying-Huan Chen, [Yu-Lun Liu](#), Brian Curless,
ACM SIGGRAPH Asia 2025 Conference Papers, 2025 [i](#) [paper](#) [i](#) [project](#)
- NeurIPS 2025 **FIPER: Generalizable Factorized Fields for Joint Image Compression and Super-Resolution.**
Yang-Che Sun, Cheng Yu Yeo, Ernie Chu, Jun-Cheng Chen, [Yu-Lun Liu](#)
Conference on Neural Information Processing Systems, 2025 [i](#) [paper](#) [i](#) [project](#)
- ICCV Workshop 2025 **Beyond Memorization: A Multi-Modal Ordinal Regression Benchmark to Expose Popularity Bias in Vision-Language Models.**
Li-Zhong Szu-Tu*, Ting-Lin Wu*, Chia-Jui Chang, He Syu, [Yu-Lun Liu](#)
The 4th DataCV Workshop and Challenge, 2025 [i](#) [paper](#) [i](#) [project](#)
- ICCV 2025 **LightsOut: Diffusion-based Outpainting for Enhanced Lens Flare Removal.**
Shr-Ruei Tsai, Wei-Cheng Chang, Jie-Ying Lee, Chih-Hai Su, [Yu-Lun Liu](#)
Proceedings of the IEEE/CVF Conference on Computer Vision, 2025 [i](#) [paper](#) [i](#) [project](#)
- ICCV 2025 **StealthAttack: Robust 3D Gaussian Splatting Poisoning via Density-Guided Illusions.**
Bo-Hsu Ke, You-Zhe Xie, [Yu-Lun Liu](#), Wei-Chen Chiu
Proceedings of the IEEE/CVF Conference on Computer Vision, 2025 [i](#) [paper](#) [i](#) [project](#)
- ICCV 2025 **LongSplat: Robust Unposed 3D Gaussian Splatting for Casual Long Videos.**
Chin-Yang Lin, Cheng Sun, Fu-En Yang, Min-Hung Chen, Yen-Yu Lin, [Yu-Lun Liu](#)
Proceedings of the IEEE/CVF Conference on Computer Vision, 2025 [i](#) [paper](#) [i](#) [project](#)
- ICCV 2025 **OpenM3D: Open Vocabulary Multi-view Indoor 3D Object Detection without Human Annotations.**
Peng-Hao Hsu, Ke Zhang, Fu-En Wang, Tao Tu, Ming-Feng Li, [Yu-Lun Liu](#), Albert Y. C. Chen, Min Sun, Cheng-Hao Kuo
Proceedings of the IEEE/CVF Conference on Computer Vision, 2025 [i](#) [paper](#)
- CoRL 2025 **See, Point, Fly: A Learning-Free VLM Framework for Universal Unmanned Aerial Navigation.**
Chih Yao Hu*, Yang-Sen Lin*, Yuna Lee, Chih-Hai Su, Jie-Ying Lee, Shr-Ruei Tsai, Chin-Yang Lin, Kuan-Wen Chen, Tsung-Wei Ke, [Yu-Lun Liu](#)
9th Annual Conference on Robot Learning, 2025 [i](#) [paper](#) [i](#) [project](#)
- CVPR 2025 **GCC: Generative Color Constancy via Diffusing a Color Checker.**
Chen-Wei Chang, Cheng-De Fan, Chia-Che Chang, Yi-Chen Lo, Yu-Chee Tseng, Jiun-Long Huang, [Yu-Lun Liu](#)
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025 [i](#) [paper](#) [i](#) [project](#)
- CVPR 2025 **AuraFusion360: Augmented Unseen Region Alignment for Reference-based 360° Unbounded Scene Inpainting.**
Chung-Ho Wu*, Yang-Jung Chen*, Ying-Huan Chen, Jie-Ying Lee, Bo-Hsu Ke, Chun-Wei Tuan Mu, Yi-Chuan Huang, Chin-Yang Lin, Min-Hung Chen, Yen-Yu Lin, [Yu-Lun Liu](#)
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025 [i](#) [paper](#) [i](#) [project](#)
- CVPR 2025 **SpectroMotion: Dynamic 3D Reconstruction of Specular Scenes.**
Cheng-De Fan, Chen-Wei Chang, Yi-Ruei Liu, Jie-Ying Lee, Jiun-Long Huang, Yu-Chee Tseng, [Yu-Lun Liu](#)
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025 [i](#) [paper](#) [i](#) [project](#)
- CVPR 2025 **FrugalNeRF: Fast Convergence for Few-shot Novel View Synthesis without Learned Priors.**
Chin-Yang Lin*, Chung-Ho Wu*, Chang-Han Yeh, Shih-Han Yen, Cheng Sun, [Yu-Lun Liu](#)
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025 [i](#) [paper](#) [i](#) [project](#)
- CVPR 2025 **DeNVer: Deformable Neural Vessel Representations for Unsupervised Video Vessel Segmentation.**
Chun-Hung Wu, Shih-Hong Chen, Chih-Yao Hu, Hsin-Yu Wu, Kai-Hsin Chen, Yu-You Chen, Chih-Hai Su, Chih-Kuo Lee, [Yu-Lun Liu](#)
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025 [i](#) [paper](#) [i](#) [project](#)
- ICLR 2025 **CAT-3DGS: A Context-Adaptive Triplane Approach to Rate-Distortion-Optimized 3DGS Compression.**
Yu-Ting Zhan*, Cheng-Yuan Ho*, Hebi Yang, Jui-Chiu Chiang, [Yu-Lun Liu](#), Wen-Hsiao Peng
International Conference on Learning Representations, 2025 [i](#) [paper](#)

- ICASSP 2025 **3D Gaussian Splatting with Grouped Uncertainty for Unconstrained Images.**
Hao-Yu Hou*, Chia-Chi Hsu*, Yu-Chen Huang*, Mu-Yi Shen*, Wei-Fang Sun, Cheng Sun, Chia-Che Chang, [Yu-Lun Liu](#), Chun-Yi Lee
IEEE International Conference on Acoustics, Speech, and Signal Processing, 2025 [i](#) [paper](#)
- WACV 2025 **ORFormer: Occlusion-Robust Transformer for Accurate Facial Landmark Detection.**
Jui-Che Chiang, Hou-Ning Hu, Bo-Syuan Hou, Chia-Yu Tseng, [Yu-Lun Liu](#), Min-Hung Chen, Yen-Yu Lin
Proceedings of IEEE/CVF Winter Conference on Applications of Computer Vision, 2025 [i](#) [paper](#) [i](#) [project](#)
Oral Presentation
- WACV 2025 **CorrFill: Enhancing Faithfulness in Reference-based Inpainting with Correspondence Guidance in Diffusion Models.**
Kuan-Hung Liu, Cheng-Kun Yang, Min-Hung Chen, [Yu-Lun Liu](#), Yen-Yu Lin
Proceedings of IEEE/CVF Winter Conference on Applications of Computer Vision, 2025 [i](#) [paper](#) [i](#) [project](#)
Oral Presentation
- NeurIPS 2024 **ReF-LDM: A Latent Diffusion Model for Reference-based Face Image Restoration.**
Chi-Wei Hsiao, [Yu-Lun Liu](#), Cheng-Kun Yang, Sheng-Po Kuo, Yucheun Kevin Jou, Chia-Ping Chen
Conference on Neural Information Processing Systems, 2024 [i](#) [paper](#) [i](#) [project](#)
- NeurIPS 2024 **Depth Anywhere: Enhancing 360 Monocular Depth Estimation via Perspective Distillation and Unlabeled Data Augmentation.**
Ning-Hsu Wang, [Yu-Lun Liu](#)
Conference on Neural Information Processing Systems, 2024 [i](#) [paper](#) [i](#) [project](#)
- NeurIPS 2024 **NaRCan: Natural Refined Canonical Image with Integration of Diffusion Prior for Video Editing.**
Ting-Hsuan Chen, Jiewen Chan, Hau-Shiang Shiu, Shih-Han Yen, Chang-Han Yeh, [Yu-Lun Liu](#)
Conference on Neural Information Processing Systems, 2024 [i](#) [paper](#) [i](#) [project](#)
- IROS 2024 **Precise Pick-and-Place using Score-Based Diffusion Networks.**
Shih-Wei Guo, Tsu-Ching Hsiao, [Yu-Lun Liu](#), Chun-Yi Lee
IEEE/RSJ International Conference on Intelligent Robots and Systems, 2024 [i](#) [paper](#) [i](#) [project](#)
- ECCV 2024 **GenRC: 3D Indoor Scene Generation from Sparse Image Collections.**
Ming-Feng Li, Yueh-Feng Ku, Hong-Xuan Yen, Chi Liu, [Yu-Lun Liu](#), Albert Y. C. Chen, Cheng-Hao Kuo, Min Sun
Proceedings of European Conference on Computer Vision, 2024 [i](#) [paper](#) [i](#) [project](#)
- ICRA Workshop 2024 **DriveEnv-NeRF: Exploration of A NeRF-Based Autonomous Driving Environment for Real-World Performance Validation.**
Mu-Yi Shen, Chia-Chi Hsu, Hao-Yu Hou, Yu-Chen Huang, Wei-Fang Sun, Chia-Che Chang, [Yu-Lun Liu](#), Chun-Yi Lee
IEEE International Conference on Robotics and Automation RoboNeRF Workshop, 2024 [i](#) [paper](#) [i](#) [project](#)
- SIGGRAPH 2024 **Matting by Generation.**
Zhixiang Wang, Baiang Li, Jian Wang, [Yu-Lun Liu](#), Jinwei Gu, Yung-Yu Chuang, Shin'ichi Satoh
ACM SIGGRAPH 2024 Conference Papers, 2024 [i](#) [paper](#) [i](#) [project](#)
- SIGGRAPH 2024 **BoostMVSNeRFs: Boosting MVS-based NeRFs to Generalizable View Synthesis in Large-scale Scenes.**
Chih-Hai Su*, Chih-Yao Hu*, Shr-Ruei Tsai*, Jie-Ying Lee*, Chin-Yang Lin, [Yu-Lun Liu](#)
ACM SIGGRAPH 2024 Conference Papers, 2024 [i](#) [paper](#) [i](#) [project](#)
- CVPR 2024 **Image-Text Co-Decomposition for Text-Supervised Semantic Segmentation.**
Ji-Jia Wu, Andy Chia-Hao Chang, Chieh-Yu Chuang, Chun-Pei Chen, [Yu-Lun Liu](#), Min-Hung Chen, Hou-Ning Hu, Yung-Yu Chuang, Yen-Yu Lin
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024 [i](#) [paper](#)
- CVPR 2024 **HumanNeRF-SE: A Simple yet Effective Approach to Animate HumanNeRF with Diverse Poses.**
Caoyuan Ma, [Yu-Lun Liu](#), Zhixiang Wang, Wu Liu, Xinchun Liu, Zheng Wang
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024 [i](#) [paper](#) [i](#) [project](#)
- ICLR 2024 **Dual Associated Encoder for Face Restoration.**
Yu-Ju Tsai, [Yu-Lun Liu](#), Lu Qi, Kelvin C.K. Chan, Ming-Hsuan Yang
The Twelfth International Conference on Learning Representations, 2024 [i](#) [paper](#) [i](#) [project](#)
- AAAI 2024 **Improving Robustness for Joint Optimization of Camera Poses and Decomposed Low-Rank Tensorial Radiance Fields.**
Bo-Yu Cheng, Wei-Chen Chiu, [Yu-Lun Liu](#)
Proceedings of AAAI Conference on Artificial Intelligence, 2024 [i](#) [paper](#) [i](#) [project](#)

- ICCV 2023 **Learning Continuous Exposure Value Representations for Single-Image HDR Reconstruction.**
Su-Kai Chen, Hung-Lin Yen, [Yu-Lun Liu](#), Min-Hung Chen, Hou-Ning Hu, Wen-Hsiao Peng and Yen-Yu Lin
Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023 [i](#) [project](#)
- ICCV 2023 **ImGeoNet: Image-induced Geometry-aware Voxel Representation for Multi-view 3D Object Detection.**
Tao Tu, Shun-Po Chuang, [Yu-Lun Liu](#), Cheng Sun, Ke Zhang, Donna Roy, Cheng-Hao Kuo, Min Sun
Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023 [i](#) [project](#)
- CVPR 2023 **Progressively Optimized Local Radiance Fields for Robust View Synthesis.**
Andreas Meuleman, [Yu-Lun Liu](#), Chen Gao, Jia-Bin Huang, Changil Kim, Min H. Kim, Johannes Kopf
Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023 [i](#) [paper](#) [i](#) [project](#)
- CVPR 2023 **Robust Dynamic Radiance Fields.**
[Yu-Lun Liu](#), Chen Gao, Andreas Meuleman, Hung-Yu Tseng, Ayush Saraf, Changil Kim, Yung-Yu Chuang, Johannes Kopf, Jia-Bin Huang
Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023 [i](#) [paper](#) [i](#) [project](#)
- ICLR 2022 **Denoising Likelihood Score Matching for Conditional Score-based Data Generation .**
Chen-Hao Chao, Wei-Fang Sun, Bo-Wun Cheng, Yi-Chen Lo, Chia-Che Chang, [Yu-Lun Liu](#), Yu-Lin Chang, Chia-Ping Chen, and Chun-Yi Lee
International Conference on Learning Representations, 2022 [i](#) [paper](#)
- ICCV 2021 **Bridging Unsupervised and Supervised Depth from Focus via All-in-Focus Supervision.**
Ning-Hsu Wang, Ren Wang, [Yu-Lun Liu](#), Yu-Hao Huang, Yu-Lin Chang, Chia-Ping Chen, and Kevin Jou
Proceedings of IEEE/CVF Conference on Computer Vision, 2021 [i](#) [paper](#) [i](#) [project](#)
- ICCV 2021 **Hybrid Neural Fusion for Full-frame Video Stabilization.**
[Yu-Lun Liu](#), Wei-Sheng Lai, Ming-Hsuan Yang, Yung-Yu Chuang, and Jia-Bin Huang
Proceedings of IEEE/CVF Conference on Computer Vision, 2021 [i](#) [paper](#) [i](#) [project](#)
- ICPR 2020 **Explorable Tone Mapping Operators.**
Chien-Chuan Su, Ren Wang, Hung-Jin Lin, [Yu-Lun Liu](#), Chia-Ping Chen, Yu-Lin Chang, and Soo-Chang Pei
Proceedings of the 25th International Conference on Pattern Recognition, 2020 [i](#) [paper](#)
- ECCV 2020 **Learning Camera-Aware Noise Models.**
Ke-Chi Chang, Ren Wang, Hung-Jin Lin, [Yu-Lun Liu](#), Chia-Ping Chen, Yu-Lin Chang, and Hwann-Tzong Chen
Proceedings of European Conference on Computer Vision, 2020 [i](#) [paper](#) [i](#) [project](#)
- CVPR 2020 **Learning to See Through Obstructions.**
[Yu-Lun Liu](#), Wei-Sheng Lai, Ming-Hsuan Yang, Yung-Yu Chuang, and Jia-Bin Huang
Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020 [i](#) [paper](#) [i](#) [project](#)
- CVPR 2020 **Single-Image HDR Reconstruction by Learning to Reverse the Camera Pipeline.**
[Yu-Lun Liu](#)^{*}, Wei-Sheng Lai^{*}, Yu-Sheng Chen, Yi-Lung Kao, Ming-Hsuan Yang, Yung-Yu Chuang, and Jia-Bin Huang
Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020 [i](#) [paper](#) [i](#) [project](#)
- AAAI 2020 **Attention-based View Selection Networks for Light-field Disparity Estimation.**
Yu-Ju Tsai, [Yu-Lun Liu](#), Yung-Yu Chuang, and Ming Ouhyoung
Proceedings of AAAI Conference on Artificial Intelligence, 2020 [i](#) [paper](#)
- AAAI 2019 **Deep Video Frame Interpolation using Cyclic Frame Generation.**
[Yu-Lun Liu](#), Yi-Tung Liao, Yen-Yu Lin, and Yung-Yu Chuang
Proceedings of AAAI Conference on Artificial Intelligence, 2019 [i](#) [paper](#) [i](#) [project](#)
Oral Presentation
- APSIPA 2014 **Background modeling using depth information.**
[Yu-Lun Liu](#) and Hsueh-Ming Hang
Signal and Information Processing Association Annual Summit and Conference, 2014
- PCS 2013 **Virtual view synthesis using backward depth warping algorithm.**
Du-Hsiu Li, Hsueh-Ming Hang, and [Yu-Lun Liu](#)
Picture Coding Symposium, 2013
[Preprints](#)
- arXiv 2026 **BRDFusion: Physics Meets Generation for Urban Scene Inverse Rendering.**
Yi-Ruei Liu, Jie-Ying Lee, Zheng-Hui Huang, [Yu-Lun Liu](#)[†], Chih-Hao Lin[†]
arXiv, 2026 [i](#) [paper](#) [i](#) [project](#)

- arXiv 2026 **Reroute, Don't Remove: Recoverable Visual Token Routing for Vision-Language Models.**
Cheng-Yu Yang, Shao-Yuan Lo, [Yu-Lun Liu](#)
arXiv, 2026 [i](#) paper [i](#) project
- arXiv 2026 **Generative World Renderer.**
Zheng-Hui Huang, Zhixiang Wang, Jiaming Tan, Ruihan Yu, Yidan Zhang, Bo Zheng, [Yu-Lun Liu](#), Yung-Yu Chuang, Kaipeng Zhang
arXiv, 2026 [i](#) paper [i](#) project
- arXiv 2026 **3AM: 3segment Anything with Geometric Consistency in Videos.**
Yang-Che Sun, Cheng Sun, Chin-Yang Lin, Fu-En Yang, Min-Hung Chen, Yen-Yu Lin, [Yu-Lun Liu](#)
arXiv, 2026 [i](#) paper [i](#) project
- arXiv 2025 **GaMO: Geometry-aware Multi-view Diffusion Outpainting for Sparse-View 3D Reconstruction.**
Yi-Chuan Huang, Hao-Jen Chien, Chin-Yang Lin, Ying-Huan Chen, [Yu-Lun Liu](#)
arXiv, 2025 [i](#) paper [i](#) project
- arXiv 2025 **Generative Refocusing: Flexible Defocus Control from a Single Image.**
Chun-Wei Tuan Mu, Jia-Bin Huang, [Yu-Lun Liu](#)
arXiv, 2025 [i](#) paper [i](#) project
- arXiv 2024 **DiffIR2VR-Zero: Zero-Shot Video Restoration with Diffusion-based Image Restoration Models.**
Chang-Han Yeh, Chin-Yang Lin, Zhixiang Wang, Chi-Wei Hsiao, Ting-Hsuan Chen, [Yu-Lun Liu](#)
arXiv, 2024 [i](#) paper [i](#) project

Talks

- Invited Talk **3D in the Wild: What to Trust, What to Reject, and What to Imagine.**
Wild3D Workshop, ECCV 2026, Malmö, Sweden, Sep. 2026.
- Invited Talk **Generative AI for Immersive 3D Worlds.**
Beyond Reality: VR × Generative AI Workshop, NYCU, Hsinchu, Taiwan, June 2026.
- Invited Talk **From Casual Capture to Deployable 3D: Reconstruction & Understanding with 3D Foundation Models.**
NVIDIA GTC Taipei 2026 Research Summit, Taipei, Taiwan, June 2026.
- Invited Seminar **Breaking Through 3DGS: Tackling Dynamics, Editing, Long Videos, and Security.**
Department of Electrical Engineering, NTHU, Hsinchu, Taiwan, Nov. 2025.
- Guest Lecture **3D from Multi-view.**
MediaTek Inc., Hsinchu, Taiwan, July 2025.
- Invited Talk **What School and Work Didn't Teach Me.**
Library, NYCU, Hsinchu, Taiwan, Sep. 2024.
- Invited Seminar **Enhancing 3D Scene Reconstruction in NeRFs.**
CSIE, NTU, Taipei, Taiwan, Mar. 2024.
- Keynote **AI Photographer: How Machines Are Changing the Way We See the World.**
Fun AI Winter Camp, NCKU, Tainan, Taiwan, Feb. 2024.
- Invited Talk **Dynamic and Local Radiance Fields for Robust View Synthesis.**
MediaTek Inc., Hsinchu, Taiwan, May. 2023.
- Invited Talk **Dynamic and Local Radiance Fields for Robust View Synthesis.**
Institute of Communications Engineering, NTHU, Hsinchu, Taiwan, April. 2023.
- Invited Talk **Dynamic and Local Radiance Fields for Robust View Synthesis.**
The 5th Augmented Intelligence and Interaction (AI²) Workshop, Taoyuan, Taiwan, March. 2023.
- Guest Lecture **What School and the Workplace Didn't Teach Me.**
NYCU, Hsinchu, Taiwan, March. 2023.
- Invited Talk **Learning to See the Unseen.**
Taichung Municipal Taichung First Senior High School, Taichung, Taiwan, March. 2023.
- Invited Talk **Learning to See the Unseen.**
Research Center for Information Technology Innovation, Academia Sinica, Taipei, Taiwan, January. 2023.
- Invited Talk **Learning to See the Unseen.**
Institute of Information Systems and Applications, NTHU, Hsinchu, Taiwan, November. 2022.

- Guest Lecture **Recent Progress on Video Restoration.**
CommE7005, NTU, Taipei, Taiwan, Mar. 2022.
- Guest Lecture **Rethinking Multimedia and IC Design in the Era of AI.**
CommE7005, NTU, Taipei, Taiwan, Nov. 2021.
- Guest Lecture **CV Final Project - Video Frame Interpolation.**
EEE5053, NTU, Taipei, Taiwan, May. 2021.
- Guest Lecture **Rethinking Multimedia and IC Design in the Era of AI.**
DEE4592, NCTU, Hsinchu, Taiwan, Mar. 2021.
- Invited Talk **Learning to See Through Obstructions.**
2020 CVPR Taiwan Sharing Session, NTU, Taipei, Taiwan, July. 2020.
- Invited Talk **Single-Image HDR Reconstruction by Learning to Reverse the Camera Pipeline.**
2020 CVPR Taiwan Sharing Session, NTU, Taipei, Taiwan, July. 2020.
- Guest Lecture **CV Final Project - Depth Estimation.**
EEE5053, NTU, Taipei, Taiwan, Jan. 2020.
- Invited Talk **Deep Video Frame Interpolation using Cyclic Frame Generation.**
The 3rd Augmented Intelligence and Interaction (AII) Workshop, Taoyuan, Taiwan, July. 2019.
- Oral **Deep Video Frame Interpolation using Cyclic Frame Generation.**
AAAI, Honolulu, Hawaii, USA, Jan. 2019.
- Guest Lecture **CV Final Project - Depth Estimation.**
EEE5053, NTU, Taipei, Taiwan, Jan. 2019.

Academic Services

- Publication Chair ACCV 2024
- COI Coordinator SIGGRAPH Asia 2025
- Area Chair / Senior Program Committee
IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2027
Asian Conference on Computer Vision (**ACCV**), 2026
Conference on Neural Information Processing Systems (**NeurIPS**), 2025, 2026
International Conference on Learning Representations (**ICLR**), 2027
AAAI Conference on Artificial Intelligence (**AAAI**), 2027
International Conference on Machine Vision Applications (**MVA**), 2025
- International Program Committee
Pacific Graphics 2025
- Organizer
2020 CVPR Taiwan Sharing Session
2025 CVPR Taiwan Night
2026 SIGGRAPH Taiwan Night
SIGGRAPH Asia 2026 Beyond Rigid Bodies Workshop
- Thesis Committees
Jin-Ting He (advisor: Yen-Yu Lin) - PhD, expected 2027
Hsuan-Kung Yang (advisor: Chun-Yi Lee) - PhD, expected 2026
Shih Fang Chen (advisor: Yen-Yu Lin) - PhD, Jun 2026
Chi-Pin Huang (advisor: Yu-Chiang Frank Wang) - PhD, Dec 2025
Yi-Hsin Chen (advisor: Wen-Hsiao Peng) - PhD, Jul 2025
Ardian Umam (advisors: Yen-Yu Lin and Jen-Hui Chuang) - PhD, Dec 2024
Cheng Sun (advisors: Hwann-Tzong Chen and Min Sun) - PhD, Jul 2023
Hitika Tiwari (advisors: K. S. Venkatesh and Yong-Sheng Chen) - PhD, Jul 2023
Cheng-Kun Yang (advisors: Yung-Yu Chuang and Yen Yu-Lin) - PhD, Jul 2023

Journal Reviewer	IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) IEEE Transactions on Image Processing (TIP) Pattern Recognition (PR) International Journal of Computer Vision (IJCV) Computer Vision and Image Understanding (CVIU) Expert Systems with Applications (ESWA) Neurocomputing Applied Soft Computing Information Fusion Transactions on Machine Learning Research (TMLR) ACM Transactions on Graphics (ACM TOG) IEEE Transactions on Visualization and Computer Graphics (TVCG) Computer Graphics Forum Computers & Graphics IEEE Transactions on Circuits and Systems for Video Technology (TCSVT) IEEE Transactions on Multimedia (TMM) Multimedia Systems The Journal of Supercomputing Transactions on Geoscience and Remote Sensing (TGRS) ACM Computing Surveys (CSUR)
Conference Reviewer / Program Committee	IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2022, 2023, 2024, 2025 IEEE Conference on Computer Vision (ICCV), 2021, 2023, 2025 European Conference on Computer Vision (ECCV), 2020, 2022, 2024 British Machine Vision Conference (BMVC), 2023, 2024 Asian Conference on Computer Vision (ACCV), 2020, 2022, 2024 Conference on Neural Information Processing Systems (NeurIPS), 2022, 2023, 2024 International Conference on Learning Representations (ICLR), 2022, 2023, 2024, 2025 International Conference on Machine Learning (ICML), 2023, 2024 AAAI Conference on Artificial Intelligence (AAAI), 2021, 2022, 2023, 2024, 2025 International Joint Conference on Artificial Intelligence (IJCAI), 2021, 2022, 2023, 2024, 2025 Conference on Robot Learning (CoRL), 2025, 2026 ACM SIGGRAPH (SIGGRAPH), 2022, 2023, 2024, 2025 ACM SIGGRAPH Asia (SIGGRAPH Asia), 2024, 2025 ACM International Conference on Multimedia (MM), 2018 IEEE International Conference on Multimedia and Expo Workshops (ICMEW), 2023 International Conference on Robotics and Automation (ICRA), 2023
Research Mentor	Research Mentor for interns at MediaTek Inc., 2019, 2020, 2021, 2022.
Professional Societies	IEEE member ACM member AAAI member TAAI member IPPR member

Teaching Experiences

Fall 2026	CSIC30191 Image and Video Generations, CS, NYCU.
Spring 2026	CSCS10017 Signals and Systems, CS, NYCU.
Fall 2025	CSIC30191 Image and Video Generations, CS, NYCU.
Fall 2025	CSIC30107 Video Compression, CS, NYCU.
Spring 2025	CSCS10017 Signals and Systems, CS, NYCU.
Fall 2024	CSIC30107 Video Compression, CS, NYCU.
Spring 2024	CSCS10017 Signals and Systems, CS, NYCU.
Fall 2023	CSIC30107 Video Compression, CS, NYCU.
Spring 2023	CSIC30107 Video Compression, CS, NYCU.

References

- Ph.D. Advisor **Yung-Yu Chuang**, *Professor of Department of Computer Science and Information Engineering*, National Taiwan University.
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- M.S. Advisor **Hsueh-Ming Hang**, *Professor of Department of Electronics Engineering*, National Chiao Tung University.
✉ hmhang@mail.nctu.edu.tw [i](#) [homepage](#)
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- Manager of Department **Chia-Ping Chen**, *Manager of the Intelligent Vision Processing Department*, MediaTek Inc.
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- Research Mentor **Wei-Sheng Lai**, *Software Engineer*, Google.
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- Research Mentor **Chen Gao**, *Research Scientist*, Meta.
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